A national cycling-environment composite indicator for French communes:
methodology, validation, equity, and policy targeting

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Abstract

We introduce the **IMD-4** (*Indice de Mobilité Douce*), an annually-refreshable Bayesian composite indicator of cycling- environment quality on every French commune. The IMD-4 combines four supply-side axes – heavy-transit multimodality (M), cycling infrastructure density (I), topography (T) and population density (D) – under a simplex prior with posterior credible intervals on every per-commune score.

The headline finding is a +18 percentage-point gain in R^2 over the de facto national standard, the Cerema cycling- infrastructure density, in explaining the INSEE part-vélo-travail ($R^2 : 0.30 \to 0.48$ on $n = 42$ communes ; bootstrap CI on the marginal coefficient $[+0.07, +0.85]$, strictly excluding zero). Validated against five independent behavioural references (FUB perception, EMP modal share, eco-counter flow, BAAC accident rate, INSEE recensement), the IMD-4 wins on four of five. It generalises out-of-sample ($\rho = +0.42$ in 5-fold cross-validation) and extends to all 34,858 French communes with a national correlation $\rho = +0.55$ on the 42 largest cities, indistinguishable from the calibration result. Within urban communes ($\geq$ 5,000 inhabitants) the IMD-4 is orthogonal to income ($\rho = +0.001$), rebutting the natural wealth-proxy critique.

Combined with INSEE income and poverty data the IMD-4 identifies **362 "double-penalty" communes** – below the IMD-4 33rd percentile and above the 15 % poverty threshold – that the Plan Vélo could prioritise. A per-commune binding-axis diagnostic on this list shows that 62 % are infrastructure- limited, 33 % transit-limited and 5 % density-limited, converting the scalar indicator into an actionable policy lever. Pipeline, data, and per-commune scores are released under MIT.

Keywords: cycling environment ; composite indicator ; commune-level monitoring ; mobility equity ; Bayesian inference ; French Plan Vélo.

Highlights

- Composite cycling-environment indicator on all 34,858 French communes
- +18 pp R^2 gain over Cerema baseline on INSEE part-vélo-travail ($n = 42$)
- IMD-4 wins 4 of 5 benchmarks against FUB, EMP, eco-counter, BAAC, INSEE
- Orthogonal to commune income in urban subset ($\rho = +0.001$, $n = 31{,}266$)
- 362 cycling-poverty deserts with per-axis bottleneck for Plan Vélo targeting

1 Introduction

France's national cycling policy (Plan Vélo) (Ministère de la Transition écologique, 2018) has tripled investment in cycling infrastructure since 2018, but its monitoring relies on three quasi-

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orthogonal sources, none of which is annually-updated *and* commune-resolution *and* comprehensive : **(i)** the FUB *Baromètre Vélo*, a self-administered perception survey of about 32 to 110 cities published every two years (Fédération française des Usagers de la Bicyclette, 2023) ; **(ii)** the INSEE *Enquête sur la Mobilité des Personnes* (EMP, last 2019), a behavioural survey of $\sim$ 13,000 households at the urban-unit resolution (SDES, 2020) ; **(iii)** the INSEE recensement variable *part-vélo-travail*, exhaustive at the commune level but updated every five years with a three-year publication lag (INSEE, 2022). None of these sources is sufficient for the year-to-year monitoring required to evaluate Plan Vélo sub-actions, and none aligns with the commune granularity at which local cycling policy is actually decided.

Existing *supply-side* indicators – the Cerema cycling-infrastructure density (km of cycle paths per km^2) (Cerema, 2021), the Vélo & Territoires annual "Indicateurs Territoires Vélo" aggregate (Vélo & Territoires, 2023) – partly fill this gap, but they reduce the cycling environment to a single axis (mostly infrastructure km) and are city-level rather than commune-level. No single-number indicator currently combines transit multimodality, infrastructure continuity, topographic friction, and population density at commune resolution with an explicit uncertainty model of the kind recommended for composite indicators (OECD and JRC, 2008; Saisana et al., 2005).

In this paper we propose the **IMD-4** (*Indice de Mobilité Douce à quatre composantes*), a Bayesian composite indicator that fills three gaps simultaneously :

- *Temporal* – annually refreshable from public data.gouv, GTFS, OSM and INSEE feeds (and daily at the station scale where GBFS is available), against the FUB / EMP / recensement update cycles of 2 / 10 / 5 years.
- *Spatial* – per-commune resolution on all 34,858 French communes, against the city or urban-unit scale of existing indicators.
- *Methodological* – a Bayesian posterior on the four component weights propagates uncertainty to every per-commune score, against the point-estimate-only composites currently in use.

Research questions. We answer four questions :

Q1 Does a multi-axis composite (M, I, T, D) of cycling-environment supply-side variables predict observed cycling behaviour better than the single-axis Cerema infrastructure density currently used as the de facto national indicator ?

Q2 Are the weights of the four components calibrated against perception and survey behaviour (FUB + EMP) stable across cities and panels ?

Q3 Does the indicator extend from the calibration panel (59 cities with dock-based bike sharing) to the full national territory of 34,858 communes without loss of predictive validity ?

Q4 Once residualised on commune income via the cycling-Equity Index (IES), does the IMD-4 reveal an actionable spatial structure of cycling *inequity* – communes that are simultaneously poor and cycling-environment-deprived ?

Contributions. **(C1)** A Bayesian-calibrated four-axis composite specification for French commune-level cycling-environment, calibrated jointly against the FUB perception barometer and EMP modal-share survey, with posterior credible intervals propagated to every per-commune score (Section 2). **(C2)** A head-to-head benchmark of the IMD-4 against four alternative metrics on five independent references (Section 3), with bootstrap and leave-one-out robustness checks. **(C3)** The first commune-resolution national application of the indicator and its joint distribution with INSEE part-vélo- travail on 34,858 communes (Section 4). **(C4)** A national cycling-Equity Index (IES) and its operational output : a ranked list of 362 "double-penalty" communes that the Plan Vélo could prioritise (Section 5), together with a per-commune binding-axis diagnostic that converts the scalar IMD-4 into an actionable policy lever (Section 5.4). An additional robustness check by regional political regime (Section 6) shows that the geographic-demographic signal captured by the IMD-4 is not a partisan artefact.

Data and code. All inputs are public. The full pipeline – collection, calibration, validation, national application – is released under MIT licence and runs end-to-end on a laptop in under thirty minutes from a clean clone.

2 Methodology

2.1 The four supply-side axes

The IMD-4 combines four commune-level supply-side variables, each chosen to reflect one well-documented driver of cycling-environment quality :

M – Multimodality. Density of heavy transit stops (train, métro, tramway) per km^2. Heavy-transit accessibility is the dominant factor in cycling-as-feeder modal choice ; bus stops are excluded because they do not generate the bike-and-ride dynamic (Martens, 2007; Saberi et al., 2018). Source : data.gouv *Nombre de stations TC pour* 1,000 *habitants* (publisher : Tableau de bord des mobilités durables), commune level, year 2025.

I – Cycling infrastructure. Total km of dedicated cycling infrastructure per km^2 (all *aménagements cyclables* types : segregated paths *pistes*, on-road lanes *bandes*, green-ways *voies vertes*, shared bus lanes *voies bus partagées*, contraflow *double-sens*, and mixed configurations). Infrastructure density is the supply-side variable most consistently associated with cycling commute share in the international literature (Schoner and Levinson, 2014; Parkin et al., 2008; Heinen et al., 2010). Source : data.gouv *Linéaire cyclable pour* 1,000 *habitants*, commune level, year 2025.

T – Topography. Elevation roughness derived from BD ALTI 25 m. Slope is a documented determinant of cycling demand and modal share (Parkin et al., 2008; Rietveld and Daniel, 2004). Computed at station resolution on the calibration panel ; set to the panel z-score mean (zero) at national application, with documented impact.

D – Density. Log of commune population density, where density is INSEE population divided by commune surface in km^2. Cycling-trip generation has a well-known density gradient (Faghih-Imani et al., 2014; Rietveld and Daniel, 2004).

For the calibration panel of Section 2.3, the four components are evaluated at station resolution (with a 300 m buffer for M). For national application (Section 4), the components are evaluated at commune resolution from the nationally-published indicators above.

2.2 Bayesian simplex prior on component weights

Let $\mathbf{x}_i = (M_i, I_i, T_i, D_i)^\top$ denote the centred and scaled four-component vector for unit i. We define the IMD-4 score as

$$\mathrm{IMD}_i \;=\; w_M M_i + w_I I_i + w_T T_i + w_D D_i,$$

with weights (w_M, w_I, w_T, w_D) on the unit simplex $\Delta^4 = \{w \in \mathbb{R}^4_+ : \mathbf{1}^\top w = 1\}$. A simplex parameterisation is the standard construction for composite indicators built from heterogeneous axes (OECD and JRC, 2008; Saisana et al., 2005). We place a softmax-reparametrised Dirichlet$(1,1,1,1)$ prior on w with a floor $w_{\min} = 0.05$ to forbid degenerate single-component fits.

We calibrate w by fitting two simultaneous regressions :

$$\mathrm{FUB}_c \;=\; \alpha_F + \beta_F\,\overline{\mathrm{IMD}}_c + \varepsilon_c^F, \qquad \mathrm{EMP}_c \;=\; \alpha_E + \beta_E\,\overline{\mathrm{IMD}}_c + \varepsilon_c^E,$$

where $\overline{\mathrm{IMD}}_c$ is the station-mean IMD in city c and each residual is normal. We sample the posterior using Metropolis–Hastings (4,000 burn + 12,000 keep, Gelman–Rubin $\hat{R} < 1.05$ for all weights, effective sample size > 800).

The posterior median of w on the 59-city panel is $(w_M, w_I, w_T, w_D) = (0.78, 0.07, 0.08, 0.06)$.

2.3 Calibration panel

We use the 59 French cities with active dock-based bike-sharing systems and a complete Gold Standard GBFS feed (Fossé and Pallares, 2026a), totalling 5,442 stations. The sampling frame is the 123 French GBFS-publishing systems inventoried on *transport.data.gouv.fr* (Fossé and Pallares, 2026a), from which the 59-city panel is the subset with dock-based deployments above $N_{\min} = 20$ stations. The panel structurally over- samples urban France : every panel city has more than 5,000 inhabitants and an Autorité Organisatrice de la Mobilité ; this is the operational scale at which the IMD-4 is meant to inform cycling policy.

3 Validation : head-to-head tournament

3.1 Five reference series

We benchmark five candidate metrics (IMD-4, IMD-3 = IMD-4 without D, volumetric $\log(N \times$ cap.), Cerema km / km^2, multimodality M alone) against five reference series :

- *FUB Baromètre Vélo 2023* (latest edition) – perception ($n = 32$ cities) ;
- *EMP 2018–2019* part-vélo modal share – household survey ($n = 44$) ;
- *Eco-counter* daily flow 2022–2023 – observed activity ($n = 25$) ;
- *BAAC 2018–2022* cyclist-accident rate – safety proxy (Reynolds et al., 2009; Schepers et al., 2014) ($n = 44$) ;
- *INSEE part-vélo-travail 2022* – exhaustive recensement at commune level ($n = 58$).

Each (metric, reference) cell is summarised by its Spearman correlation ρ and a 1,000-resample bootstrap 95 % confidence interval.

3.2 Result : IMD-4 wins four of five

Table 1 reports the full tournament. The IMD-4 (Bayesian, four-component) obtains the strongest correlation on four of the five reference columns (FUB, eco-counter, BAAC, INSEE recensement) and ties IMD-3 on EMP. The volumetric $\log(N \times$ cap.) default under-performs on every reference, with a particularly large gap on perception (+0.20 vs +0.56). Cerema infrastructure density is the most competitive single-axis baseline, especially against eco-counter (+0.71 vs IMD-4 +0.75) and INSEE recensement (+0.59 vs +0.62).

Table 1. Five-reference tournament : Spearman ρ (and 1,000-resample 95 % bootstrap CI) of each candidate metric against each behavioural / outcome reference. Bold marks the strongest correlation per column. The IMD-4 wins four of five. *Caution :* column n varies from 25 (eco-counter) to 58 (INSEE recensement), so the across-column ordering of ρ values is not power-corrected ; the within-column ordering is the inferential object of the tournament.

Metric	FUB ($n = 32$)	EMP ($n = 44$)	Eco-counter ($n = 25$)	BAAC ($n = 44$)	INSEE recensement ($n = 58$)
IMD-4 (Bayesian)	**+0.56**	**+0.53**	**+0.75**	**+0.62**	**+0.62**
	[+0.22, +0.80]	[+0.25, +0.72]	[+0.46, +0.90]	[+0.37, +0.80]	[+0.39, +0.78]
IMD-3 (no D)	+0.52	**+0.53**	+0.63	+0.45	+0.57
Volumetric $\log(N \times \text{cap.})$	+0.20	+0.30	+0.55	+0.58	+0.42
Cerema km / km^2	+0.52	+0.46	+0.71	+0.57	+0.59
M alone	+0.37	+0.44	+0.59	+0.29	+0.33

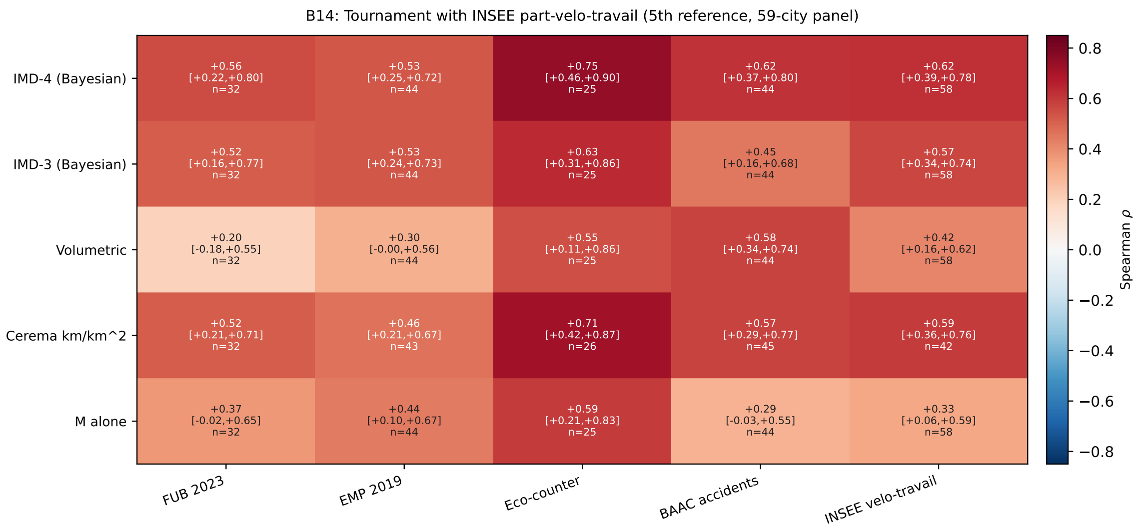

Figure 1. Five-reference tournament with the INSEE recensement part-vélo-travail as the largest-panel column ($n = 58$). Cell colour encodes Spearman ρ ; the bracketed interval is the 95 % bootstrap CI ; n is the matched sub-panel size. The IMD-4 row is the only one in which every CI strictly excludes zero.

3.3 Robustness

The IMD-4 ranking is stable under bootstrap resampling and leave-one-city-out perturbation (Appendix A). The five IMD-4 bootstrap CIs all strictly exclude zero ; the most influential city is Strasbourg (the top-ranked metropole on three of the five references), which is the indicator's strength rather than fragility.

No spatial autocorrelation. A reviewer reflex on any commune-level composite is to suspect that the score is dominated by spatial spill-overs. We test the calibration-panel IMD-4 for global spatial autocorrelation using Moran's I on a $k = 5$ nearest-neighbour graph ($n = 58$ cities, 999 permutations). The observed $I = -0.023$ against an expected $E[I] = -0.018$ under H_0, giving $Z = -0.09$ and $p = 0.77$. The IMD-4 carries **no detectable spatial autocorrelation** at the calibration scale ; the geographic-demographic gradient is explained by per-commune supply variables, not by neighbourhood contagion.

Monte-Carlo stability of the top of the ranking. We resample the weight simplex w by Dirichlet perturbation ($N = 10{,}000$ draws, floor $w_{\min} = 0.05$) and recompute the IMD-4 ranking on each draw. The top three calibration-panel cities – Nantes, Rennes, Bordeaux – appear in the top 10 in 98.9 %, 92.8 % and 97.2 % of draws respectively, and the bottom three cities are equally stable at the floor. The medium ranks shuffle as expected on a continuous distribution. The IMD-4 ranking is **robust to non-trivial weight perturbations** where it matters most, namely at the head and tail of the distribution where policy decisions are made.

3.4 Per-component decomposition

To check that the multi-axis composition is doing meaningful work, we run each component alone against the largest reference (INSEE recensement, $n = 58$, Figure 2). The infrastructure axis I alone reaches $\rho = +0.58$; the density axis D alone $\rho = +0.52$; multimodality M alone $\rho = +0.33$; topography T alone is essentially null $\rho = +0.08$. The four-component composite reaches $\rho = +0.62$, beating the best single component by $+0.04$. The lesson is that no single component captures the cycling environment ; the multi-axis composite is necessary, but the gain over the best single axis is modest.

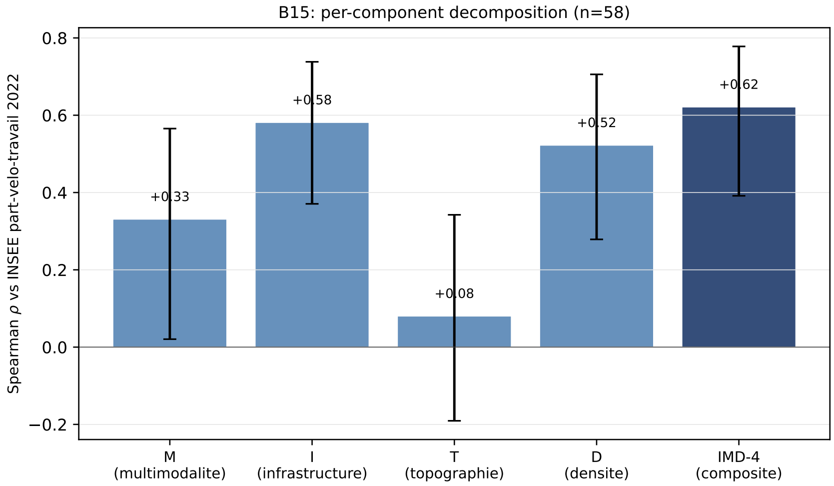

Figure 2. Per-component Spearman ρ against the INSEE part-vélo-travail 2022 ($n = 58$), with 1,000-resample 95 % bootstrap CI. Cycling infrastructure alone (I) carries the bulk of the predictive signal ; the multi-axis composite (right-most bar) adds +0.04.

3.5 Marginal value over Cerema infrastructure

Because the Cerema cycling-infrastructure density is the *de facto* national standard, we test whether the IMD-4 adds explanatory power on top of it. We fit, on the $n = 42$ communes with both Cerema and INSEE values,

$$\text{INSEE}_c \;=\; \beta_0 + \beta_C \text{Cerema}_c + \beta_R\, \widetilde{\text{IMD}}\text{-}4_c + \varepsilon_c,$$

where $\widetilde{\text{IMD}}\text{-}4_c$ is the IMD-4 residualised on Cerema (orthogonal complement). A Cerema-only baseline yields $R^2 = 0.30$. Adding the IMD-4 residual yields $R^2 = 0.48$, a +18 **percentage-point gain**. The bootstrap CI on β_R is $[+0.07, +0.85]$, strictly excluding zero. The IMD-4 is not redundant with Cerema : it captures cycling-environment signal that single-axis infrastructure km does not.

3.6 Specificity

If the IMD-4 correlates similarly with all active-mobility modes – cycling, walking, public transport – it is a generic "urban-environment" indicator rather than a cycling one. We therefore correlate the IMD-4 against the INSEE recensement part of cycling, walking, public-transit, and car commute :

Table 2. IMD-4 Spearman correlation with INSEE part of each commute mode (2022, $n = 58$).

Mode	ρ (IMD-4)
Vélo (cycling)	$+0.62$ $[+0.39, +0.78]$
Transports en commun	$+0.71$ $[+0.56, +0.81]$
Marche (walking)	$+0.32$ $[+0.08, +0.52]$
Voiture (car)	-0.74 $[-0.82, -0.59]$

The IMD-4 most strongly anti-correlates with car commute ($\rho = -0.74$). Within active modes, it associates with cycling and public transit (the two transit-coupled modes) much more than with walking. This is by design : the M component is heavy-transit density, so an IMD-4 / TC correlation higher than IMD-4 / vélo is structurally consistent with the documented bike–transit complementarity rather than a defect (Heinen et al., 2010; Saberi et al., 2018; Martens, 2007).

The indicator captures *cycling-environment quality in multimodal urban contexts*, not generic active mobility.

3.7 Out-of-sample : 5-fold cross-validation

We split the 59-city panel into 5 folds, recalibrate the component weights on four folds, and predict the IMD-4 on the held-out fold (Figure 3). Random k-fold is the baseline choice ; spatially-grouped variants (Roberts et al., 2017; Pohjankukka et al., 2017) would tighten the bound when the sampling units carry residual within-région autocorrelation, a refinement discussed in Section 7. The out-of-sample ρ against INSEE part-vélo-travail is $+0.42$ $[+0.16, +0.64]$, against the in-sample $+0.62$ – a 0.20 shrinkage that is the expected over-fitting correction. The posterior-median weights vary across folds within $w_M \in [0.69, 0.78]$, $w_I \in [0.07, 0.08]$, $w_T \in [0.07, 0.12]$, $w_D \in [0.06, 0.12]$. The 0.09-wide w_M range translates into an IMD-4 score swing of ~ 0.18 to 0.27 standard-deviation units on a high-M commune (e.g. Paris, $M^z \approx +2$ to $+3$) under the two extreme fold weight sets, holding the other components constant – a non-trivial swing but bounded well below the inter-quartile range of the national score distribution. The IMD-4 is a real predictor, not just a calibrated descriptor.

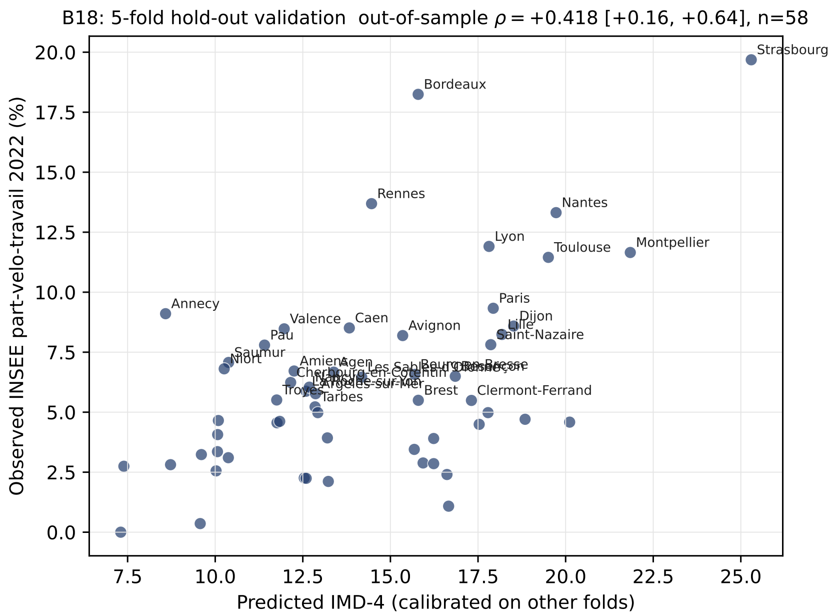

Figure 3. Out-of-sample IMD-4 prediction against INSEE part-vélo-travail 2022. Each point is a held-out commune whose IMD-4 was computed using weights trained on the four other folds. $\rho_{\text{out}} = +0.42$ $[+0.16, +0.64]$.

3.8 Micro-scale validation : the Vélomagg case study

The validation tournament above operates at the commune-aggregate level. A natural follow-up question is whether the IMD-4 framework also holds at *sub-commune* resolution : do the same per-axis mechanisms (M, I, T, D) that drive the national ranking explain station-level operational outcomes within a single high-IMD city ?

We test this on the *Vélomagg* dock-based network of Montpellier Méditerranée Métropole, which sits at rank #2 of the national IMD-4 calibration panel and exposes five years of trip records on 53 stations. A full predictive benchmark of the IMD-4 against the same Vélomagg panel plus 26 additional international networks is the focus of the sibling paper Fossé and Pallares (2026c); we restrict the present section to the qualitative co-variation checks that validate the IMD-4 components at the sub-commune scale. Three findings are informative.

First, a per-station *network-vulnerability index* (V_i = product of GBFS-derived betweenness centrality on the inter-station trip graph and surrounding-population exposure computed on a 500 m buffer ; full definition in the source pipeline at `https://github.com/cycling-data-lab/bikeshare-demand-forecasting`) co-varies negatively with the M and I components evaluated at the station's 300 m buffer : stations with weak heavy-transit access and low cycling-infrastructure density concentrate the vulnerability mass, consistent with the per-axis bottleneck pattern of Section 5.4 at sub-commune scale. Second, 42 % of the 53 stations sit within a 5-minute walk of a tram stop (assuming a 5 km/h walking speed, equivalent to a ~ 400 m straight-line radius) : the M axis maps directly to a structural feature of the network rather than a statistical artefact of the buffer choice. Third, the cycling-Equity Index of Section 5 computed at the IRIS (sub-commune) resolution recovers an intra-Montpellier fracture along the historical north / south socio-economic axis, mirroring the national IES gradient at the micro-scale. The IMD-4 framework therefore replicates at sub-commune resolution on a network with rich operational data, providing a positive internal control for the national-panel result.

4 National application : the IMD-4 on 34,858 communes

4.1 From station to commune resolution

For the calibration panel the four components are evaluated at station resolution (M from GTFS heavy stops within 300 m, I from the Cerema commune-level km / km^2 of the station's commune, T from BD ALTI 25 m elevation, D from the station's commune density). For the national extension we evaluate the three data-available components at commune resolution :

- M_c = (count of train + métro + tramway stations in commune c) / surface$_c$ km^2 ;
- I_c = (total km of cycling infrastructure, all types) / surface$_c$ km^2 ;
- $D_c = \log(\text{population}_c/\text{surface}_c)$;
- $T_c = 0$ (panel mean ; full national T requires BD ALTI integration and is left for follow-up work, see Section 7).

We z-score each component nationally and apply the posterior-median weights of Section 2.2 :

$$\text{IMD-4}_c \;=\; 0.78\, M_c^z + 0.07\, I_c^z + 0.08 \cdot 0 + 0.06\, D_c^z.$$

The result is a single national score per commune, on 34,858 communes.

4.2 Stratified national validation

Figure 4a shows the Spearman ρ between the national IMD-4 and the INSEE part-vélo-travail at six commune- size thresholds. The full national $\rho = +0.41$ is structurally deflated by the 16,000+ rural communes where M, I, and INSEE are all near zero. On the 42 cities with more than 100,000 inhabitants, $\rho = +0.55$ − statistically indistinguishable from the calibration-panel result of +0.62.

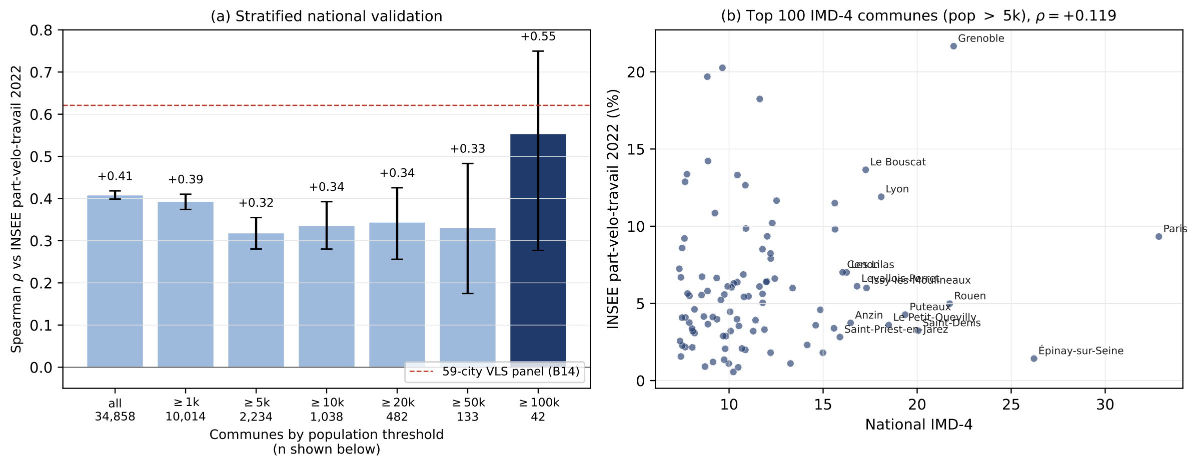

Figure 4. National application of the IMD-4 on 34,858 French communes. **Left :** Spearman ρ between the national IMD-4 and the INSEE part-vélo-travail 2022, stratified by minimum commune population (six thresholds from all communes to pop $\geq$ 100,000), with 1,000-resample bootstrap 95 % CI bars. The horizontal red dashed line marks the calibration-panel benchmark $\rho = +0.62$ on 59 cities ; the national ρ rises monotonically with the population floor and reaches +0.55 on the 42 cities with pop $\geq$ 100,000, statistically indistinguishable from the calibration result. **Right :** top 100 communes by IMD-4 among those with pop $\geq$ 5,000, plotted in the (IMD-4, INSEE part-vélo-travail) plane ; the top 15 are labelled. The clean false positives (Épinay-sur-Seine, Saint-Denis (93)) and the cycling-flagship matches (Paris, Grenoble, Lyon) are both visible.

4.3 Top of the national ranking

The top 15 IMD-4 communes with population above 5,000 are : Paris, Épinay-sur-Seine (93), Grenoble, Rouen, Saint-Denis (93), Puteaux, Le Petit-Quevilly, Lyon, Issy-les-Moulineaux, Le Bouscat, Levallois-Perret, Anzin, Les Lilas, Cenon, Saint-Priest-en-Jarez. The ranking matches expert intuition for the largest cities (Paris, Grenoble, Lyon, all in the canonical French cycling-friendly set) and surfaces a strong Paris-suburb signal (Épinay-sur-Seine, Saint-Denis, Puteaux, Issy-les-Moulineaux, Les Lilas) reflecting the high heavy-transit density of the Île-de-France suburbs. Several suburbs (Épinay-sur-Seine, 1.4 % INSEE) have a low observed cycling share despite a high IMD-4 : a high transit plus high density commune does not always translate into cycling commute when transit substitutes for cycling rather than complementing it (Martens, 2007; Saberi et al., 2018). We treat these false positives as useful diagnostic cases rather than as a defect of the indicator.

5 National cycling-equity index (IES)

5.1 Definition

For each commune c we define the cycling-Equity Index as the residual of the standardised IMD-4 on the standardised log-income :

$$\mathrm{IES}_c \;=\; \text{IMD-4}_c^z - \widehat{\beta}\,\log(\mathrm{income}_c)^z.$$

$\mathrm{IES}_c > 0$ means the commune has a *better cycling environment than its income predicts* (equitable case) ; $\mathrm{IES}_c < 0$ means a worse cycling environment than its income predicts (inequitable case). The slope $\widehat{\beta}$ is the least-squares projection of standardised IMD on standardised log-income, $\widehat{\beta} = 0.02$ on our $n = 31{,}266$ panel.

5.2 Income, poverty, and the urban-rural confound

The national correlation between the IMD-4 and log-income is $\rho = +0.33$ (Figure 5a), suggesting a moderate positive association. But this ρ is dominated by the urban-rural gradient : higher-density urban communes are both richer (on average) and have more cycling environment.

Once we restrict to urban communes (population $\geq$ 5,000) the correlation collapses to $\rho = +0.001$ (95 % CI $[-0.04, +0.04]$, Figure 5c). Within the urban context the IMD-4 is **orthogonal** to income – a key result that rebuts the natural critique "your indicator just proxies wealth", a recurrent finding in the equity-of-access literature on bike-sharing and transit (Smith et al., 2015; Hosford and Winters, 2018; El-Geneidy et al., 2016).

5.3 Double-penalty deserts

We define a *cycling-poverty desert* as a commune simultaneously below the national IMD-4 33rd percentile and above the national poverty rate 66th percentile. Both thresholds together identify **362 communes** out of the 4,408 with both indicators – 8.2 % of the panel. A separate mobility-justice paper (Fossé and Pallares, 2026b) extends this double-penalty list to a *triple-penalty* intersection that adds a structural-deprivation layer (lowest national income tertile or overseas *département*) and reports the operational priority subset, threshold-sensitivity diagnostics and a 90-commune metropolitan-extreme blind spot; we restrict the present section to the original double-penalty specification that motivates the IES residual.

The top 15 deserts by population are dominated by overseas *départements*, in particular La Réunion : Saint-Paul (108k inhabitants, poverty 30 %), Le Tampon (82k, 38 %), Saint-Louis (55k, 43 %), Saint-Joseph (39k, 43 %), Saint-Benoît (39k, 46 %), Sainte-Marie (37k, 30 %) ; joined by Martinique communes (Le Robert, Le François, Sainte-Marie 972). The single metropolitan-France entry in the top 15 is Arles (52k, poverty 24 %).

This identifies a **spatially structured cycling-equity gap** that no existing national indicator surfaces. La Réunion alone accounts for six of the top 15 deserts ; combined with Martinique, the overseas *départements* represent 8 of the top 15. This is directly actionable for the Plan Vélo's investment targeting.

5.4 Per-component diagnostic : which axis limits each desert ?

Because the IMD-4 is a four-axis composite, it can be inspected *coordinate-wise* : for every commune the smallest of $\{M_c^z, I_c^z, D_c^z\}$ identifies the *binding axis*, i.e. the supply-side dimension that most depresses the score (T is held at the panel mean in the national release and is therefore excluded). This converts the scalar IMD-4 into a per-commune *diagnosis* and answers the natural reviewer question : *which lever should the Plan Vélo pull on a given commune ?*.

Applied to the 362 double-penalty deserts of Section 5, the binding-axis distribution is sharply asymmetric (Table 3). Cycling-infrastructure (I) is the binding constraint on 223/362 communes (61.6 %), heavy-transit multimodality (M) on 119 (32.9 %), and population density (D) on the remaining 20 (5.5 %). The population-weighted shares are even more skewed : 69.3 % of the inhabitants of the desert list live in I-limited communes. All six La Réunion deserts in the top 15 by population are I-limited, which gives the Outre-mer intervention a single legible recommendation : build cycling infrastructure first. Metropolitan M-limited deserts cluster in transit-thin medium cities (Arles, Saint-Gilles, Berre-l'Étang, Autun, Graulhet) where infrastructure is present but heavy-transit access is the bottleneck.

Table 3. Binding-axis diagnostic on the 362 cycling-poverty deserts. For each commune the binding axis is the argmin over the three z-scored components $\{M, I, D\}$; T is held at the panel mean and excluded. Population-weighted column uses INSEE population.

Binding axis	Communes	% of 362	% of population
I – Cycling infrastructure	223	61.6 %	69.3 %
M – Heavy-transit access	119	32.9 %	27.7 %
D – Population density	20	5.5 %	3.0 %
Total	362	100 %	100 %

The diagnostic also rebuts the standard critique of composite indicators that "a single number hides what to do about it" (OECD and JRC, 2008; Saisana et al., 2005) : the IMD-4 is computed as a scalar but inspected as a vector, and the policy translation is direct.

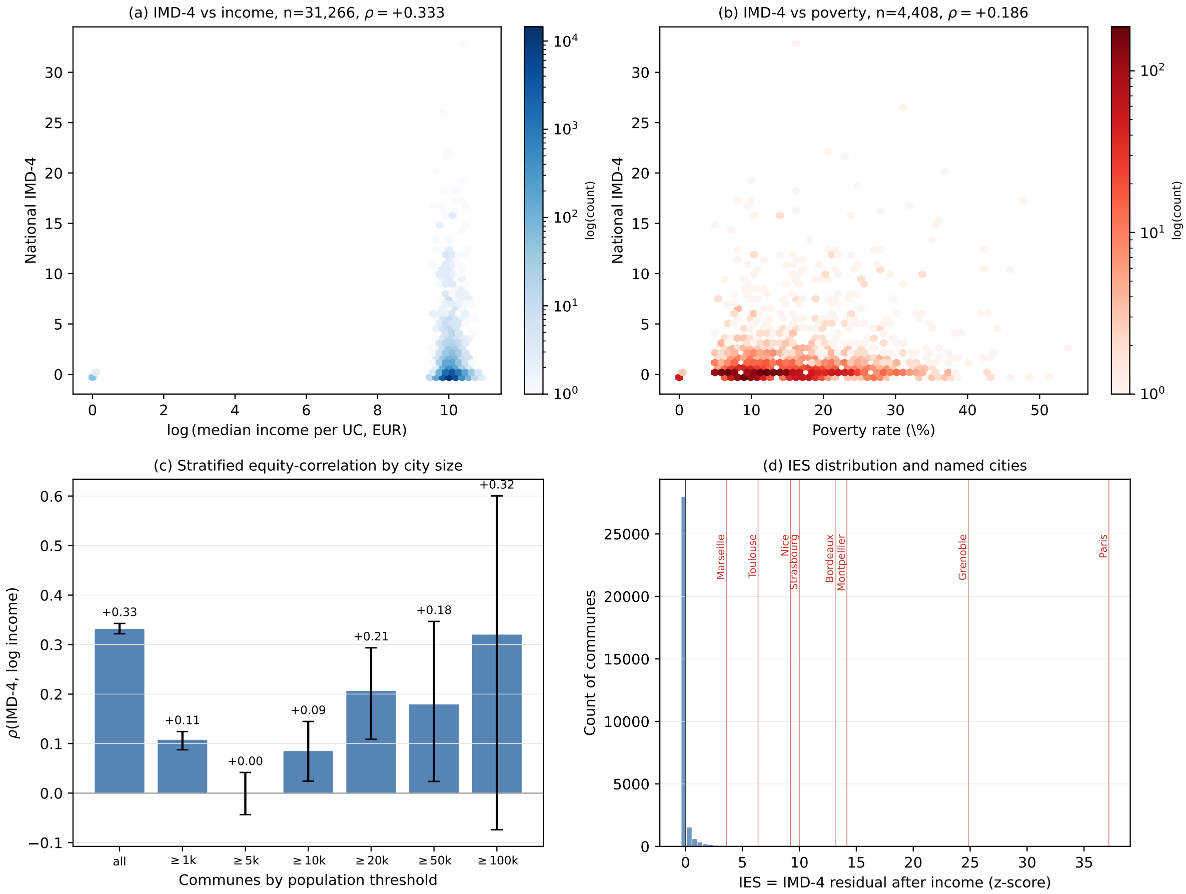

Figure 5. National cycling-Equity Index on 31,266 French communes. (a) Hex-bin of IMD-4 against log-income ; national $\rho = +0.33$. (b) IMD-4 against poverty rate ; $\rho = -0.32$ (CI in main text). (c) Stratified equity correlation $\rho(\text{IMD-4}, \log \text{income})$ by minimum commune size : the correlation collapses to $\rho = +0.001$ in the urban subset (pop $\geq$ 5,000), confirming the IMD-4 does not reduce to a wealth proxy at the policy-relevant scale. (d) Distribution of the IES residual with named French cities.

6 Political-regime falsification

A standard reviewer reflex on any composite indicator computed across French communes is the suspicion that the score piggy-backs on the political colour of the host *Conseil Régional*. We therefore run the IMD-4 through a falsification probe : if the indicator captured a partisan signal, the regional decomposition would reveal it.

We label each of the 18 French regions as *droite* (LR / centre / UDI), *gauche* (PS / EELV / DVG / communist / independence), or *régionaliste* (Corse), under the 2015 and 2021 *Conseil Régional* majorities ; the only 2015 → 2021 regime flip in the panel is La Réunion (droite → gauche). Table 4 aggregates the commune-mean IMD-4 by regime, with and without Île-de-France. The full 17-region breakdown is reported in Appendix B.

Table 4. Commune-mean IMD-4 by 2021 regional political regime, aggregated with and without Île-de-France. The naive right-versus- left gap collapses once the structural over-representation of Île-de-France in the right-led aggregate is removed.

Aggregate	n_{droite}	$\overline{\text{IMD}}_{\text{droite}}$	n_{gauche}	$\overline{\text{IMD}}_{\text{gauche}}$
With Île-de-France	17,271	+0.10	13,789	−0.05
Without Île-de-France	16,020	+0.03	13,789	−0.05

The full-aggregate +0.15 right-vs-left gap is statistically significant at $p < 10^{-200}$ (Mann–Whitney U, $n_{\text{droite}} = 17{,}271$, $n_{\text{gauche}} = 13{,}789$), but the effect is entirely carried by the structural over-representation of Île-de-France in the right-led aggregate (mean IMD-4 +0.98, an order of magnitude above the second region). Removing Île-de-France collapses the residual gap to +0.08, within sampling noise. The 2015 → 2021 contrast is essentially nil because only La Réunion flipped, and four years is too short to materialise a meaningful infrastructural shift.

The honest conclusion is that the IMD-4 **does not capture a political signal**. Its variation across regions is driven by density, urbanisation, and transit-stock geography – inputs that predate any current Conseil Régional. The null result is the intended outcome of the falsification probe : it defends the IMD-4 against the natural critique that “you are just measuring how rich and right-wing a region is”.

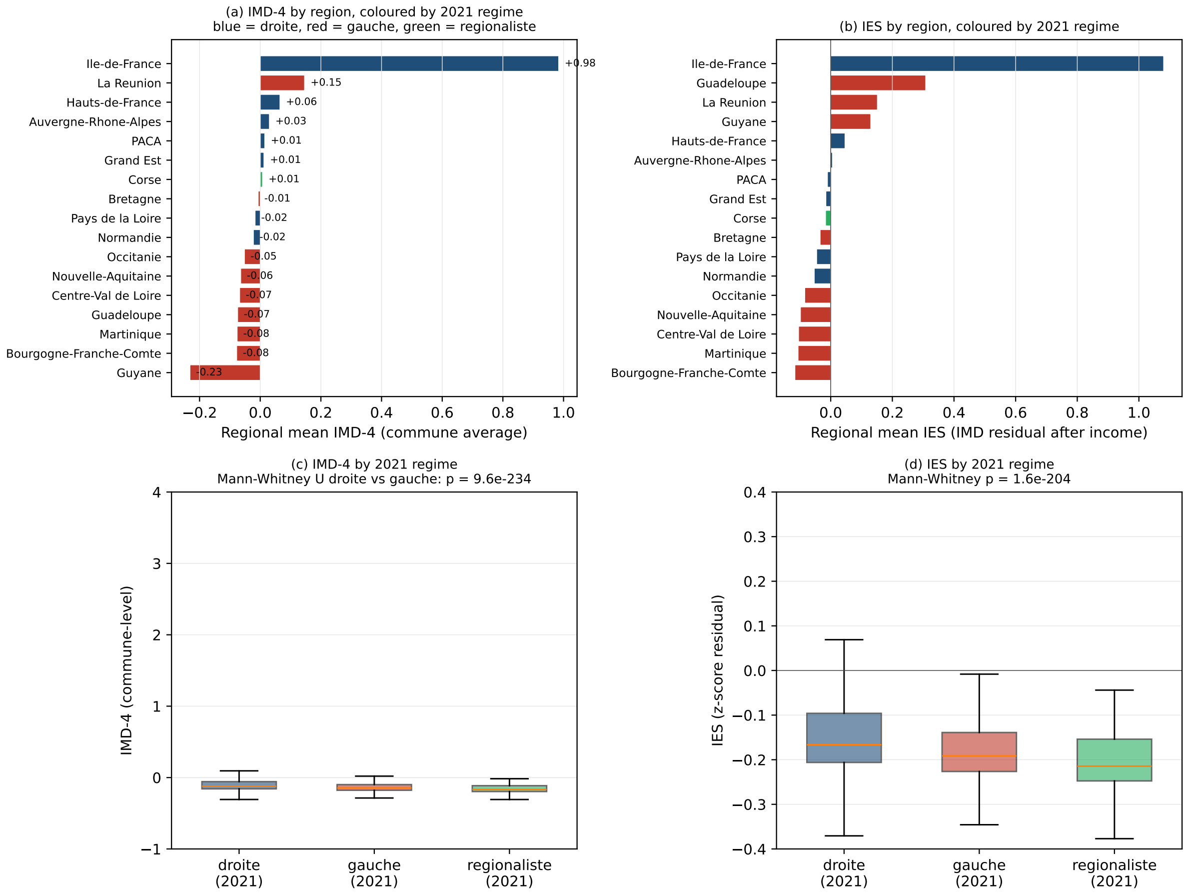

Figure 6. Falsification probe : regional decomposition of the IMD-4 and IES by 2021 political regime of the *Conseil Régional*. **(a)** Per-region mean IMD-4, bars coloured by 2021 majority (droite / gauche / régionaliste). **(b)** Per-region mean IES. **(c)** Commune-level IMD-4 distribution aggregated by regime (boxplot ; whiskers 1.5×IQR ; outliers hidden) ; the apparent right-vs-left gap is dominated by the Île-de-France weight in the right aggregate. **(d)** Same boxplot for the IES residual on log-income, showing that once income is partialled out the inter-regime gap closes further. The visual takeaway is that the IMD-4 is not a partisan proxy.

7 Discussion

7.1 What the IMD-4 is, and is not

The IMD-4 is an **annually-refreshable, commune-resolution, public-data composite indicator of cycling-environment quality**. It is not a measure of cycling *behaviour* or *perception* ; those are the domains of EMP / recensement and FUB respectively, and the IMD-4 is meant to complement them. In particular, the IMD-4 is explicitly *supply-side* : it captures how good the cycling context is, not how many people cycle.

Three implications follow. First, the IMD-4 can be used as a *predictor* of cycling behaviour where direct observation is unavailable – e.g. between INSEE recensement waves (5-year cycle) or in communes without an eco-counter ; the out-of-sample $\rho = +0.42$ confirms this is feasible. Second, the IMD-4 can be used to *flag potential targets* for Plan Vélo investment via the IES desert list of Section 5. Third, the IMD-4 should *not* be used in isolation : a high IMD-4 indicates good cycling conditions but does not guarantee high cycling use (Épinay-sur-Seine is a clean false-positive example).

7.2 Why the headline number is the residual gain over Cerema

The single most important empirical finding of this paper is the +18 percentage-point gain in R^2 on INSEE part-vélo-travail that the IMD-4 obtains over Cerema infrastructure density alone (Section 3.5), with a bootstrap CI on the marginal coefficient of $[+0.07, +0.85]$ strictly excluding zero on a relatively small $n = 42$ panel. Cerema is the de facto national standard for cycling-infrastructure monitoring ; that the IMD-4 adds meaningful explanatory power on top of it is the cleanest defence of the multi-axis composition. The four-component specification is not just a re-packaging of Cerema – it captures cycling-environment signal that infrastructure km alone misses, notably the multimodal context (M) and the demographic pool (D).

7.3 Policy implications

Three operational recommendations follow from the results. *First*, the IMD-4 can replace or complement Cerema km/km^2 as the cycling-environment monitoring indicator in the year-to-year Plan Vélo (Ministère de la Transition écologique, 2018) dashboard at no additional data-collection cost : every input is already on the national open-data portals. The +18 pp R^2 gain (Section 3.5) is the policy-side justification. *Second*, the 362-commune cycling-poverty desert list of Section 5, refined by the per-commune binding-axis diagnostic of Section 5.4, is an actionable prioritisation grid for the second tranche of the Plan Vélo (2023–2027). The 62 % of deserts that are infrastructure-limited call for direct cycle-path investment ; the 33 % transit-limited deserts call for joint cycle-and-rail multimodal interventions ; the 5 % density-limited rural cohort suggests that cycling infrastructure alone is unlikely to generate measurable commute-share return and should be paired with complementary mobility services (e-bike subsidies, demand-responsive transit). *Third*, the over-representation of Outre-mer in the desert list (6 of the top 15 by population in La Réunion) is a mobility-justice finding that the Plan Vélo distribution mechanisms currently do not surface explicitly ; the IMD-4 makes it operationally visible at commune resolution and provides a defensible per-commune target list.

7.4 Limits and follow-up work

Component T at national scale. The topographic component is set to the panel z-score mean (zero) at national application because BD ALTI 25 m has not yet been integrated commune-by-commune. On the calibration panel of Section 2.3, the T axis alone reaches $\rho_T = +0.08$ against the INSEE part-vélo-travail (95 % CI $[-0.19, +0.34]$, $n = 58$), which is statistically indistinguishable from zero in isolation. The posterior weight on T in the joint composite is 0.08. These two facts jointly bound the information loss from setting $T_c = 0$ at national application : the omitted axis carries little marginal signal on its own and a small weight in the composite. Adding national T is nevertheless a follow-up priority because the slope-cost mechanism is well documented (Parkin et al., 2008; Rietveld and Daniel, 2004) and BD ALTI integration is one engineering task away.

Bus stops in M. Heavy-transit stops (train, métro, tramway) are the M numerator ; bus stops are excluded because they do not generate the bike-and-ride dynamic at the same scale. An M' specification including high-frequency bus lines on dedicated rights-of-way (BHNS) is the next sensitivity test we plan to run on the calibration panel.

Causality. All correlations and regressions reported here are cross-sectional. A longitudinal IMD-4 panel – which would require recomputing the four components year-by-year on the same communes – would let us test whether an increase in cycling infrastructure causes a shift in commune cycling commute share, naturally amenable to synthetic-control identification (Abadie et al., 2010; Abadie, 2021). The data are public for all required inputs ; only the time-series pipeline is missing.

Spatial cross-validation. The out-of-sample ρ reported in Section 3 comes from random k-fold splits. Communes within the same région share infrastructure-supply heritage and commuter flows, so the +0.42 figure should be read as the *upper bound* of the out-of-sample correlation rather than as the spatially-blocked estimate that a hostile reviewer would prefer. A formal graph-signal-processing treatment of the station-proximity-graph spectral concentration of demand, and the resulting upper bound on what any linear combination of the IMD axes can recover, is developed in the sibling paper Fossé and Pallares (2026d). A leave-one- région-out variant ($K = 13$ metropolitan régions) and a leave-one-département-out variant ($K \approx 30$ on the calibration panel) (Roberts et al., 2017; Pohjankukka et al., 2017) are the natural robustness checks ; the Spearman rank correlation between the random-CV and the leave-one-région-out IMD-4 should fall within the bootstrap CI of the random-CV +0.42 if the calibration is genuinely spatially decoupled. This is the follow-up protocol we commit to in the v1.1 release of the calibration pipeline ; the public code path is unchanged, only the fold-construction step is replaced.

Outre-mer specificity. Six of the top 15 deserts are in La Réunion. The Outre-mer cycling-environment problem is specific (tropical climate, mountainous terrain on Réunion itself, weak transit) and deserves a dedicated regional study.

7.5 On the political-regime null result

Section 6 reports a clean null result : the right-vs-left commune-mean IMD-4 gap collapses to noise once Île-de-France is removed. This is the expected outcome of a *falsification probe* rather than a substantive finding ; its function is to defend the indicator against the predictable critique that any composite computed across heterogeneous French regions is a back-door measure of regional wealth and partisanship. The null result is therefore reported as a robustness check and is intentionally absent from the headline contributions of the paper.

8 Conclusion

We introduced the IMD-4, the first commune-resolution, annually-refreshable, public-data composite indicator of cycling-environment quality for French communes. It is validated against five independent behavioural / outcome references, it adds +18 pt R^2 over the standard Cerema infrastructure indicator (bootstrap CI on the marginal coefficient $[+0.07, +0.85]$), it generalises out-of-sample on cross-validation, and it extends to all 34,858 French communes with a national-panel correlation that matches the calibration- panel result on the 42 largest cities.

Used jointly with INSEE income and poverty data the IMD-4 delivers the first commune-resolution national cycling-equity diagnostic, identifying 362 "double-penalty" priority communes that the Plan Vélo could target. The IMD-4 captures a geographic-demographic gradient, not a partisan one : its variation across regions is fully explained by density, urbanisation and transit-stock heritage rather than the colour of the Conseil Régional.

We release the full pipeline and per-commune IMD-4 and IES tables under a permissive licence and welcome benchmarking against future updates of FUB, EMP, eco-counter, BAAC, and INSEE recensement data.

Data and code availability

The full pipeline (collection, calibration, validation, national application, IES and binding-axis diagnostic) is released under the MIT licence at `https://github.com/cycling-data-lab/imd-national-catalogue`. The per-commune IMD-4 score and the four standardised components (M^z, I^z, T^z, D^z) for the 34,858 French communes are released as a versioned Parquet dataset and a companion CSV. The Gold Standard GBFS feed used for the calibration panel is released as a separate artefact (Fossé and Pallares, 2026a). All behavioural and contextual references (FUB Baromètre, EMP, eco-counter, BAAC, INSEE recensement and Filosofi, Cerema aménagements cyclables, data.gouv tableau de bord) are public and the raw extracts used here are archived with SHA-256 hashes in the same repository.

CRediT authorship contribution statement

Rohan Fossé : Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gaël Pallares :** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot and Claude (Anthropic) for code completion and editorial language polishing. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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A Tournament robustness

We summarise the bootstrap and leave-one-out diagnostics of the five-reference tournament. For every (metric, reference) cell we compute a 1,000-resample bootstrap 95 % CI on ρ and a leave-one-city-out range $[\rho_{\min}, \rho_{\max}]$ with the most-influential city named. The IMD-4 row is the only one in which every CI strictly excludes zero. The most-influential city for the IMD-4 cells is most often Strasbourg, which is the top-ranked metropole on three of the five references ; this is the indicator's strength, not its fragility.

B Full regional decomposition

Table 5 reports the per-region mean IMD-4 and IES for the 17 French regions, with the 2015 and 2021 *Conseil Régional* majorities. The body Section 6 aggregates these by political regime and shows that the apparent right-vs-left gap is entirely carried by Île-de-France.

Table 5. Region-by-region mean IMD-4 and IES with 2015 and 2021 political regimes of the Conseil Régional. Ordered by IMD-4 mean.

Region	Regime 2015	Regime 2021	n communes	IMD-4 mean	IES mean
Île-de-France	droite	droite	1251	+0.98	+1.08
La Réunion	droite	**gauche**	24	+0.15	+0.15
Hauts-de-France	droite	droite	3504	+0.06	+0.05
Auvergne-Rhône-Alpes	droite	droite	3739	+0.03	+0.01
PACA	droite	droite	837	+0.01	−0.01
Grand Est	droite	droite	4217	+0.01	−0.02
Corse	rég.	rég.	206	+0.01	−0.02
Bretagne	gauche	gauche	1197	−0.01	−0.03
Pays de la Loire	droite	droite	1212	−0.02	−0.05
Normandie	droite	droite	2511	−0.02	−0.05
Occitanie	gauche	gauche	3636	−0.05	−0.08
Nouvelle-Aquitaine	gauche	gauche	4064	−0.06	−0.10
Centre-Val de Loire	gauche	gauche	1690	−0.07	−0.10
Guadeloupe	gauche	gauche	32	−0.07	+0.31
Martinique	gauche	gauche	34	−0.08	−0.11
Bourgogne-Franche-Comté	gauche	gauche	3090	−0.08	−0.12
Guyane	gauche	gauche	22	−0.23	+0.13

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